

# Unit - 3 Generative AI and Large Language Models

## What is Generative AI?

**Generative AI (Generative Artificial Intelligence)** is a type of AI that can **create new content** based on the patterns it has learned from existing data.

Traditional AI often focuses on **analyzing, classifying, or predicting** something.

Generative AI focuses on **creating something new**.

### Examples of content generated by AI

- **Text** → articles, answers, summaries, stories
- **Images** → photographs, artwork, illustrations
- **Code** → Python, Java, C++, SQL, etc.
- **Audio** → speech, music, sound
- **Video** → animations, videos, synthetic scenes

### Simple Example

If we give an AI system the prompt:

**“Write a short story about a robot living on Mars.”**

A generative AI model can create a completely new story.

Similarly:

**“Create an image of a futuristic city on Mars.”**

A generative image model can generate an image based on this description.

### Generative AI vs Traditional AI

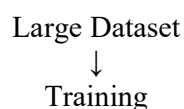
| Traditional AI                | Generative AI                                |
|-------------------------------|----------------------------------------------|
| Mainly analyzes existing data | Creates new content                          |
| Classification and prediction | Content generation                           |
| Example: Spam detection       | Example: Writing an email                    |
| Output may be a label/class   | Output can be text, image, audio, code, etc. |
| Example: “Spam/Not Spam”      | Example: Generates a reply                   |

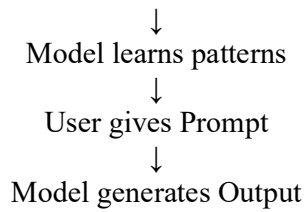
### How Does Generative AI Work?

Generative AI models are trained on **large amounts of data**.

During training, the model learns patterns, structures, relationships, and representations from the data.

#### Basic process:





### **Example:**

Suppose an AI model is trained on a large collection of text.

It learns patterns such as:

“The sun rises in the \_\_\_\_.”

The model learns that **east** is a likely continuation.

When given:

“Artificial intelligence is...”

the model predicts what text is likely to come next based on its learned patterns and the context.

## **Types of Generative AI**

Generative AI can create different types of content.

### **1. Text Generation**

Generates human-like text.

Examples:

- Chatbots
- Story generation
- Summarization
- Question answering
- Email generation

### **2. Image Generation**

Creates images from text or other inputs.

Example:

“A cat wearing sunglasses sitting on a motorcycle.”

The model generates an image based on the description.

### **3. Code Generation**

Generates computer programs or code.

Example:

“Write Python code to calculate factorial.”

The AI can generate the corresponding Python code.

#### **4. Audio Generation**

Can generate:

- Speech
- Music
- Sound effects

#### **5. Video Generation**

Can generate or transform video content from prompts or other inputs.

## **Large Language Models (LLMs)**

**Large Language Models (LLMs)** are AI models trained on very large amounts of text data to **understand and generate human-like language**.

Examples include models such as **GPT-family models, Gemini, Claude, and Llama**.

LLMs are primarily designed to work with language, although modern multimodal models can also process other types of information.

### **Example:**

User: “Explain photosynthesis in simple words.”

An LLM can generate:

“Photosynthesis is the process by which green plants use sunlight, water, and carbon dioxide to produce food and oxygen.”

The model does not simply retrieve one fixed answer. It generates a response based on patterns and representations learned during training.

### **Why are LLMs called "Large"?**

The word **Large** refers mainly to the scale of the model and its training.

LLMs generally involve:

- Very large training datasets
- Large numbers of parameters
- Significant computational resources
- Complex neural-network architectures

### **What is a Parameter?**

A **parameter** is a numerical value learned by a neural network during training.

Parameters are adjusted during training so that the model becomes better at its task.

You can think of parameters as **learned numerical settings** inside the model.

A larger number of parameters does **not automatically mean** a model is better, but large models can represent complex patterns.

## How are LLMs Trained?

This is an important examination topic.

A simplified LLM training process is:

**Data Collection → Data Preprocessing → Tokenization → Model Training → Parameter Optimization → Evaluation → Fine-tuning/Alignment → Deployment**

Let's understand each step.

### Step 1: Data Collection

A large amount of text data is collected from appropriate sources.

The data may contain:

- Books
- Articles
- Websites
- Documents
- Code
- Other text sources

The quality and composition of training data are extremely important.

### Step 2: Data Preprocessing

The collected data is cleaned and prepared.

Common operations may include:

- Removing corrupted data
- Removing unwanted content
- Deduplication
- Normalizing data
- Filtering low-quality content
- Preparing training examples

The exact preprocessing pipeline differs between models.

## Tokenization in LLMs

LLMs generally do not process an entire sentence directly as a single object.

Text is converted into **tokens**.

For example:

“Artificial intelligence is powerful.”

may be broken into tokens representing words or parts of words.

Conceptually:

Artificial | intelligence | is | powerful

Modern tokenizers often use **subword tokenization**, so a word does not necessarily correspond to exactly one token.

The tokens are then converted into numerical IDs.

Example:

Text → Tokens → Token IDs

These numerical representations can be processed by the neural network.

## Training Objective of an LLM

One of the most important ideas behind many modern LLMs is **next-token prediction**.

Suppose the training text is:

“The cat is sitting on the mat.”

The model may receive:

“The cat is sitting on the”

and learn to predict:

“**mat**”

Similarly:

“Machine learning is a branch of”

The model learns that:

“**artificial intelligence**”

is a likely continuation.

During training, the model performs this prediction repeatedly over huge amounts of text.

## Neural Network and Transformer Architecture

Modern LLMs are commonly based on the **Transformer architecture**.

The Transformer was introduced in the influential 2017 paper:

“**Attention Is All You Need.**”

The key mechanism is **self-attention**.

Self-attention helps the model determine which other tokens are important when processing a particular token.

## Example

Consider:

“The animal didn't cross the road because **it** was tired.”

To understand “**it**”, the model needs to consider the surrounding context and determine what “it” refers to.

Self-attention allows the model to assign different levels of importance to other tokens while processing the sequence.

## Self-Attention

**Self-attention** is a mechanism that allows a model to consider relationships between different tokens in the same sequence.

Conceptually:

**Input tokens** → **Attention mechanism** → **Context-aware representations**

For example:

“The bank was crowded because people were depositing money.”

The word **bank** should be understood in the context of:

“depositing money”

rather than:

“river”

Self-attention helps the model capture such relationships.

## Training an LLM: Simplified Mathematical View

During training:

1. Text is converted into tokens.
2. Tokens are converted into numerical representations.
3. The model predicts the next token.
4. The predicted probability distribution is compared with the actual next token.
5. A **loss function** measures the error.
6. **Backpropagation** calculates how parameters contributed to the error.
7. An **optimizer** updates the parameters.
8. This process is repeated many times.

## Basic flow

**Input → Prediction → Loss → Backpropagation → Parameter Update → Better Prediction**

This process is repeated across a very large training dataset.

## **Pre-training and Fine-tuning**

LLM development is commonly described using two major stages.

### **Pre-training**

The model learns general language patterns from a very large dataset.

It learns things such as:

- Vocabulary
- Grammar
- Relationships between words
- General knowledge patterns
- Code patterns, when code is included in training
- Long-range contextual relationships

The model is not manually given a grammar rule for every sentence. Much of this structure is learned from data.

### **Fine-tuning**

After pre-training, a model can be further trained on a more specific dataset for a particular task or behavior.

Examples:

- Medical question answering
- Legal text processing
- Programming
- Customer support

Modern assistant models may also undergo additional **instruction tuning and preference/alignment training** so they better follow user instructions and produce useful responses.

### **LLM Example:**

Suppose the prompt is:

**“Explain Newton's First Law in simple language.”**

The LLM processes the input as tokens.

It uses its learned parameters and contextual representations to estimate suitable next tokens.

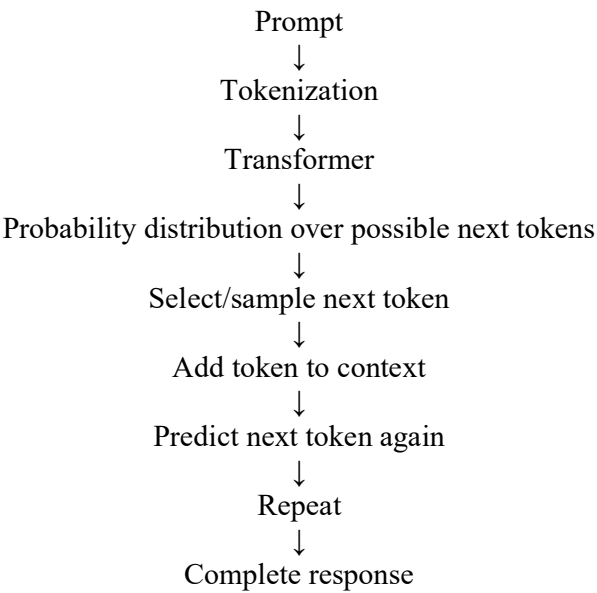
Conceptually:

Explain → Newton's → First → Law → ...

The model generates the response **token by token**, with each generated token influencing the context for subsequent generation.

**How Does an LLM Generate Text?**

At a simplified level:



For example:

“The capital of India is”

Possible probabilities might conceptually look like:

| Token     | Probability |
|-----------|-------------|
| New Delhi | High        |
| Mumbai    | Low         |
| Kolkata   | Low         |
| Chennai   | Low         |

The model selects a likely continuation and continues generating.

The actual process uses probability distributions and decoding strategies rather than simply selecting the most frequent word every time.

**LLM vs NLP**

This distinction is useful for exams.

| NLP                                                    | LLM                                                           |
|--------------------------------------------------------|---------------------------------------------------------------|
| Broad field of AI/computational linguistics            | A type of large neural language model                         |
| Includes many techniques                               | Usually based on deep learning                                |
| Includes tokenization, POS tagging, NER, parsing, etc. | Can perform many language tasks using learned representations |
| Can use traditional ML and DL                          | Usually uses Transformer-based architectures                  |
| Broader concept                                        | Specific class of models                                      |

# Generative AI vs LLM

Another important distinction:

**Generative AI is the broader concept.**

**LLMs are one important type of generative AI model for language.**

For example:

## Generative AI

- Text generation
- Image generation
- Audio generation
- Video generation
- Code generation

## LLM

- Mainly language/text
- Also can support code and, in multimodal systems, interact with other modalities

## Popular LLMs, Tokens, Context Window, Temperature and Multimodal AI

### Popular LLMs

| LLM / AI Model Family | Organization |
|-----------------------|--------------|
| GPT / ChatGPT         | OpenAI       |
| Gemini                | Google       |
| Claude                | Anthropic    |
| Llama                 | Meta         |

### GPT / ChatGPT – OpenAI

**GPT (Generative Pre-trained Transformer)** is a family of AI models developed by **OpenAI**.

**ChatGPT** is an AI assistant/product that uses OpenAI models.

Common applications:

- Text generation
- Coding
- Question answering
- Summarization
- Reasoning
- Content creation

### Gemini – Google

**Gemini** is Google's family of generative AI models.

It is designed to work with multiple types of information, depending on the model, including:

- Text
- Images
- Audio
- Video
- Code

## **Claude – Anthropic**

**Claude** is a family of AI models developed by **Anthropic**.

It is designed for tasks such as:

- Text generation
- Analysis
- Coding
- Summarization
- Question answering

## **Llama – Meta**

**Llama** is a family of large language models developed by **Meta**.

It is widely used by researchers and developers to build AI applications and services.

## **Tokens**

A **token** is a basic unit of text that an AI language model processes.

A token may represent:

- A complete word
- Part of a word
- Punctuation
- Another piece of text

The exact tokenization depends on the model and tokenizer.

### **Example:**

The sentence:

**"Artificial Intelligence is useful."**

is converted by the model into a sequence of tokens before processing.

Therefore, an LLM does not simply process text as humans see it; internally, it processes **tokens**.

## **Importance of Tokens**

Tokens are important because they affect:

1. **Input size** – how much information can be provided.

2. **Output size** – how much text can be generated.
3. **Context window** – the amount of information the model can consider.
4. **API cost** – many AI APIs measure usage in tokens.

**A token is a small unit of text processed by an LLM.**

## Context Window

The **context window** is the maximum amount of token-based information that a model can consider at one time when generating a response.

The context may include:

- User instructions
- Previous conversation
- Documents
- Input text
- Other information supplied to the model

### Example:

Suppose an AI model has a context window of **100,000 tokens**.

It can consider up to approximately 100,000 tokens of context within that model's limit.

A larger context window allows an AI system to work with:

- Long books
- Large documents
- Large codebases
- Long conversations
- Multiple documents

**Context window is the maximum amount of token-based information an AI model can consider at one time.**

## Temperature

**Temperature** is a parameter used during text generation to control the randomness or variation in the model's output.

### Low Temperature

A lower temperature generally produces:

- More predictable output
- More consistent responses
- Less variation

**Suitable for:**

- Factual answers
- Structured tasks
- Repetitive generation

## High Temperature

A higher temperature generally produces:

- More varied output
- More creative responses
- Greater randomness

**Suitable for:**

- Creative writing
- Brainstorming
- Generating different ideas

## Example

Prompt: "Write a sentence about the ocean."

With a **low temperature**, the response is more likely to be conventional and predictable.

With a **higher temperature**, the response can be more diverse and creative.

## Important Point

Temperature does **not** make an AI model more intelligent. It mainly affects the **randomness and variation of generated output**.

**Temperature controls how predictable or varied an AI model's generated response is.**

## Multimodal AI

**Multimodal AI** is an AI system that can understand and/or generate information in multiple modalities, such as:

- **Text**
- **Images**
- **Audio/Voice**
- **Video**
- **Code**

The term **multimodal** means that the AI can work with more than one type of information.

## Traditional AI vs Multimodal AI

### Traditional Text-Based AI

A text-based AI mainly works with:

**Text → AI → Text**

Example:

User types a question → AI generates a text answer.

### Multimodal AI

A multimodal AI can work across multiple modalities:

**Text + Image + Voice + Video → AI → Text/Voice/Image/etc.**

For example, a user may:

1. Upload an image.
2. Ask a question using voice.
3. Receive a text or spoken answer.

This creates a more natural human-computer interaction.

## **GPT-4o as a Multimodal AI Concept**

**GPT-4o** ("o" for **omni**) is an example of a model designed to work across multiple modalities.

The important concept is that the model is designed for interaction involving:

**Text + Image + Audio/Voice**

Instead of treating each modality as completely separate, a multimodal system can process different forms of information within a unified model architecture or integrated system.

### **Example:**

A user shows an image of a circuit and asks through voice:

"What is wrong with this circuit?"

A multimodal AI can potentially:

- **See** the circuit image.
- **Understand** the spoken question.
- **Analyze** the information.
- **Respond** in text or voice.

This is more powerful than a system that can only process text.

## **Gemini and Multimodal AI**

**Gemini** is Google's multimodal AI model family.

Depending on the specific Gemini model and product, it can work with multiple modalities such as:

- Text
- Images
- Audio
- Video
- Code

### **Example**

A student can provide:

- A photograph of a mathematical problem
- A text instruction
- A voice question

The AI can combine these inputs to understand the user's request and generate an appropriate response.

## Why Multimodal AI is Important

Multimodal AI makes interaction with computers more natural and flexible.

### Applications

#### Education

- Analyze diagrams
- Explain photographs
- Solve questions from images
- Voice-based learning

#### Healthcare

- Analyze medical images alongside other information, where appropriate

#### Robotics

- Combine visual information with language and other sensor data

#### Customer Service

- Voice + text + images

#### Content Creation

- Text, images, audio and video generation

## Quick Revision

| Concept               | Simple Meaning                                                       |
|-----------------------|----------------------------------------------------------------------|
| <b>LLM</b>            | AI model designed to understand and generate language                |
| <b>Token</b>          | Small unit of text processed by an AI model                          |
| <b>Context Window</b> | Maximum token-based context the model can consider at one time       |
| <b>Temperature</b>    | Controls randomness/variation in generated output                    |
| <b>Multimodal AI</b>  | AI that works with multiple types of information                     |
| <b>GPT-4o</b>         | OpenAI model designed for multimodal interaction                     |
| <b>Gemini</b>         | Google's family of generative AI models with multimodal capabilities |
| <b>Claude</b>         | Anthropic's family of AI models                                      |
| <b>Llama</b>          | Meta's family of large language models                               |

## AI Image Generation, Limitations of LLMs and Responsible AI Use

# AI Image Generation

**AI Image Generation** is a technology that uses Artificial Intelligence models to create images from text prompts, images, or other forms of input.

For example:

**Prompt:** "A futuristic city at night with flying cars"

An AI image-generation model can create an image based on this description.

## How AI Image Generation Works

Modern image-generation systems learn patterns from large datasets containing images and associated information.

A simplified process is:

**User Prompt → AI Model → Image Generation → Generated Image**

The user provides a **prompt**, which describes the desired image. The AI model interprets the prompt and generates an image matching the description.

## Applications

AI image generation is used in:

- Graphic design
- Advertising
- Education
- Game development
- Animation
- Marketing
- Concept art
- Product design
- Creative content

## Stable Diffusion

**Stable Diffusion** is a family of generative AI models used for creating images from text and other inputs.

### Key Features

- Text-to-image generation
- Image-to-image generation
- Image editing and transformation
- Can be used through different applications and interfaces
- Widely used by developers, researchers, and creators

### Example

**Prompt:**

"A realistic mountain landscape during sunset"

Stable Diffusion can generate an image based on this description.

## Midjourney

**Midjourney** is a generative AI system focused primarily on creating images from natural-language prompts.

### Key Features

- Text-to-image generation
- Strong focus on artistic and creative images
- Useful for concept art and visual design
- Allows users to create different visual styles from prompts

### Example

"A futuristic robot walking through a cyberpunk city"

The system generates visual interpretations of the prompt.

## DALL-E

**DALL-E** is a generative AI image model developed by **OpenAI**.

It can generate images based on natural-language descriptions.

### Key Features

- Text-to-image generation
- Creative image generation
- Can generate different concepts and compositions
- Useful for design, education, and creative applications

### Example

"A penguin wearing a space suit on Mars"

The model can create an image based on this description.

## Comparison of AI Image Generation Systems

| System           | Organization                    | Main Purpose                                    |
|------------------|---------------------------------|-------------------------------------------------|
| Stable Diffusion | Stability AI and open ecosystem | Image generation and customization              |
| Midjourney       | Midjourney                      | Artistic and creative image generation          |
| DALL-E           | OpenAI                          | Generating images from natural-language prompts |

## Limitations of LLMs

Although LLMs are powerful, they have several important limitations.

## Hallucination

**Hallucination** occurs when an AI model generates information that sounds correct but is actually false, inaccurate, or unsupported.

### Example

A user asks: "Who invented a fictional technology?"

The model may provide a confident-sounding name and explanation even though the person or technology does not exist.

## Why Hallucination Happens

LLMs generate responses by predicting likely sequences of tokens based on learned patterns. They do not automatically guarantee that every generated statement is factually correct.

## How to Reduce the Risk

- Verify important information.
- Use reliable sources.
- Check facts and references.
- Do not blindly trust confident-sounding answers.

## Simple Definition

**Hallucination is the generation of false or unsupported information by an AI model.**

## Knowledge Cutoff

A **knowledge cutoff** refers to the point in time up to which the model's training knowledge was collected.

Information created after that point may not be included in the model's original training data.

### Example:

Suppose a model's training knowledge ends in **2024**.

A user asks:

"What happened yesterday?"

The model may not know the answer from its training data unless it has access to an appropriate up-to-date information source.

## Important Point

A knowledge cutoff does **not** mean that the model cannot ever provide current information. AI systems can sometimes use tools such as web search or other external data sources to obtain more recent information.

## Simple Definition

**Knowledge cutoff** is the date or point up to which information was included in a model's training knowledge.

## Bias

**Bias** occurs when an AI system produces systematically unfair, unbalanced, or stereotyped results.

Bias can arise from:

- Training data
- Human-generated content
- Historical inequalities
- Data imbalance
- Model design
- User prompts

## Example

If training data contains strong stereotypes about a particular group, an AI model may reproduce those stereotypes in its output.

## Problems Caused by Bias

Bias can lead to:

- Unfair decisions
- Stereotyping
- Discrimination
- Misleading information
- Unequal treatment

## How to Reduce Bias

- Use diverse and representative datasets.
- Test models for unfair outcomes.
- Review AI-generated results.
- Include human oversight.
- Avoid treating AI output as automatically neutral.

## Simple Definition

**AI bias** is a systematic tendency of an AI system to produce unfair, unbalanced, or stereotypical results.

## Responsible Use of AI

**Responsible AI use** means using Artificial Intelligence in an ethical, transparent, safe, and accountable manner.

Important areas include:

1. **Academic Integrity**
2. **Plagiarism**
3. **Disclosure of AI-Generated Content**

## **Academic Integrity**

**Academic integrity** means being honest and responsible in academic work.

Students should use AI as a **learning and assistance tool**, not as a way to falsely represent someone else's or AI's work as their own.

### **Responsible Use**

Students can use AI to:

- Understand difficult concepts
- Generate practice questions
- Improve grammar
- Brainstorm ideas
- Get explanations
- Review their own work

Students should follow their institution's rules when using AI for assignments, projects, examinations, or research.

### **Example**

#### **Responsible:**

Student uses AI to understand a difficult topic and then writes the answer independently.

#### **Irresponsible:**

Student submits an AI-generated assignment as completely original work when the academic rules do not permit it.

## **Plagiarism**

**Plagiarism** is presenting another person's words, ideas, or work as one's own without proper acknowledgment.

AI-generated content can also create academic-integrity issues when a student submits it as their own work contrary to applicable rules.

### **How to Avoid Plagiarism**

- Write in your own words.
- Cite sources properly.
- Use quotation marks for direct quotations.
- Acknowledge sources and assistance when required.
- Follow your university's academic-integrity policy.
- Do not submit AI-generated work as original work if the rules prohibit it.

## **Disclosure of AI-Generated Content**

**AI disclosure** means clearly stating when AI has been used to create or significantly assist with content, when disclosure is required by the relevant institution, publisher, employer, or platform.

## Example

A student may be required to state:

"AI tools were used to assist with brainstorming and language editing."

The exact disclosure requirement depends on the rules of the university, teacher, journal, organization, or platform.

## Why Disclosure is Important

Disclosure promotes:

- Transparency
- Academic honesty
- Accountability
- Trust
- Responsible use of AI

## Quick Revision Table

| Topic                      | Meaning                                            |
|----------------------------|----------------------------------------------------|
| <b>AI Image Generation</b> | Creating images using generative AI                |
| <b>Stable Diffusion</b>    | Generative AI family used for image generation     |
| <b>Midjourney</b>          | AI system focused on creative image generation     |
| <b>DALL-E</b>              | OpenAI image-generation model                      |
| <b>Hallucination</b>       | Generation of false or unsupported information     |
| <b>Knowledge Cutoff</b>    | Point up to which training knowledge was collected |
| <b>Bias</b>                | Unfair, unbalanced, or stereotypical AI behavior   |
| <b>Academic Integrity</b>  | Honesty and responsibility in academic work        |
| <b>Plagiarism</b>          | Presenting another's work or ideas as one's own    |
| <b>AI Disclosure</b>       | Transparently stating AI use when required         |